ThinkStack

Testing Phase Document

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Chapter 1

TEST PLAN

1.1 Purpose and Scope

This document records how ThinkStack is tested, what the tests establish, and what they currently report. It covers the backend, the interface, the contract between them, and the packaged application as installed on a user's machine.

It does not yet cover cross-platform verification of the replaced interface or of the Scribe writing path. Section 5.2 states precisely what is outstanding and why, rather than leaving the omission implicit.

1.2 Testing Strategy

Four layers, each establishing something the layer below it cannot.

Table 1.1: Testing layers and what each establishes

Layer	**Tool**	**What only this layer can establish**
Unit (backend)	pytest	That a function computes what it claims, including its failure modes.
Unit (interface)	vitest, jsdom	That a component renders and behaves under interaction, without a browser.
Contract	pytest, source parsing	That both sides of the API agree, parsed from source rather than asserted by hand.
Reachability	live HTTP	That the handlers *run*. Parsing proves the paths match; only calling them proves anything executes.
Packaged application	`validate_bundle.py`	That the *installed* program works. A suite run from a source checkout cannot see what the freezer omitted.

Why the last two layers exist. Version 1.0.0 passed every test and every build, was published, and did not start: the frozen backend silently dropped the inference library's shared objects and the embedding model's data, because neither package ships a hook for the freezer. And when the interface was replaced, the contract test was found to be hardcoded to the old directory, the moment the replacement shipped it would have gone on passing while validating a tree nobody built.

1.3 Test Environment

Table 1.2: Test environment

Item	Value
Development machine	16 GB memory, 16-core x86-64 CPU, no usable accelerator
Operating system	Fedora Linux (kernel 7.1.3)
Python	3.14, dependencies pinned exactly
Node	22.x
Inference	llama.cpp via a local client; Qwen2.5 0.5B and 1.5B, Q4_K_M
Embeddings	all-MiniLM-L6-v2, local
Typesetting	Tectonic, bundled
Continuous integration	GitHub Actions, Ubuntu runners

Interface tests are run with file parallelism disabled. Vitest defaults to one worker per core; on a 16-core machine already running an editor that is sixteen Node processes each with its own DOM implementation, which exhausted memory and froze the machine. Single-threaded the same 198 tests complete in about five seconds.

Chapter 2

TEST CASES

2.1 Metadata Extraction

Table 2.1: Metadata extraction test cases

ID	Case	Expected	Result
ME-01	Largest horizontal text at the top of page 1	Exact title	Pass
ME-02	Rotated archive stamp larger than the title	Stamp ignored	Pass
ME-03	Horizontal margin furniture (journal banner)	Ignored	Pass
ME-04	Title wrapping to a second line	Both lines joined	Pass
ME-05	Same font reused lower on the page	Not joined to the title	Pass
ME-06	Small-capitals title (two sizes, one baseline)	Whole title	Pass
ME-07	Font change mid-word	No space inserted	Pass
ME-08	Ligature stored as one glyph	Expanded	Pass
ME-09	Accent stored before its letter	Recomposed	Pass
ME-10	Hyphenated title across a line break	De-hyphenated	Pass
ME-11	Authors in columns	Each cell a name	Pass
ME-12	Authors comma-separated in one cell	Split correctly	Pass
ME-13	One author per row with affiliation	Name kept, affiliation dropped	Pass
ME-14	Affiliation row naming only some employers	Whole row rejected	Pass
ME-15	Phrase repeated in the author region	Treated as an address	Pass
ME-16	Accented institution name	Still recognised	Pass
ME-17	Email in brace form	Not read as a name	Pass
ME-18	Scanned page, no text layer	Empty, not a guess	Pass

ID	Case	Expected	Result
ME-19	Year from archive identifier vs. stray number	Identifier wins	Pass
ME-20	Paper stating no year	Empty	Pass
ME-21	Implausible year	Refused	Pass
ME-22	Plausible title, no authors	Model fallback invoked	Pass
ME-23	Model returns an empty field	Good value retained	Pass
ME-24	Model disabled and result implausible	Returned unchanged	Pass

2.2 Documents, Storage and Encryption

Table 2.2: Document lifecycle and encryption test cases

ID	Case	Expected	Result
DS-01	Upload a valid PDF	Ingested, chunks stored	Pass
DS-02	Upload a non-PDF	Rejected, 400	Pass
DS-03	Upload an empty file	Rejected, 400	Pass
DS-04	PDF yielding no text	422, document not retained	Pass
DS-05	Layout extraction fails	Ingest still succeeds	Pass
DS-06	Analysis queue unavailable	Document retained, warning logged	Pass
DS-07	Rename propagates to every chunk	All chunks updated	Pass
DS-08	Rename with a blank title	400, nothing written	Pass
DS-09	Rename an absent document	404	Pass
DS-10	Delete removes chunks, file and cached analysis	All removed	Pass
DS-11	Encrypt, then read chunk text	Ciphertext stored	Pass
DS-12	Decrypt with the wrong password	Refused, detected	Pass
DS-13	PDF route for an encrypted document	403	Pass
DS-14	PDF served inline, not as an attachment	Inline	Pass

2.3 Paper Writer

Table 2.3: Paper writer test cases

ID	Case	Expected	Result
PW-01	Compile a valid document	PDF produced	Pass
PW-02	Missing package declaration	Inserted, compiles	Pass
PW-03	Bare fragment with no preamble	Wrapped, compiles	Pass
PW-04	One malformed figure	Document still produced	Pass
PW-05	Index requested	`.ind` generated in Python	Pass
PW-06	Figure referenced from a project directory	Resolves	Pass
PW-07	Filename escaping the project directory	Refused	Pass
PW-08	Symbolic link out of the project	Refused	Pass
PW-09	Rename a project	Compile, preview and paths unaffected	Pass
PW-10	Cite a paper from the editor	Entry written, key returned	Pass
PW-11	Cite the same paper twice	One entry, same key	Pass
PW-12	A key already in the bibliography	Honoured, not recomputed	Pass
PW-13	Metadata improved after a citation	Original key retained	Pass
PW-14	An entry written by hand	Retained alongside	Pass
PW-15	A key written by hand collides	Suffixed, hand entry untouched	Pass
PW-16	Listing what can be cited	Nothing written to disk	Pass
PW-17	Citing a document not in the library	404	Pass
PW-18	Citations with no bibliography declared	Declaration supplied, references typeset	Pass
PW-19	Citations with no bibliography file	Nothing added, compile unaffected	Pass
PW-20	`\citep`, `\citet`, `\nocite`, page-qualified	All recognised	Pass
PW-21	A hand-written `thebibliography`	Respected, nothing added	Pass
PW-22	`&`, `%`, `$` in a title	Escaped, file valid	Pass

ID	Case	Expected	Result
PW-23	Capitalisation in a title	Preserved against the style	Pass
PW-24	Compile a cited document end to end	Numbered references and a reference list in the PDF	Pass

2.4 References and Correction

The two cases that open this table are the reason the section exists. Neither failure raises an error: one prints the wrong names, the other prints a question mark, and both look to the user like the feature not working.

Table 2.4: Reference storage and correction test cases

ID	Case	Expected	Result
RC-01	Eight authors stored and read back	Eight names, none split	Pass
RC-02	A name written surname first	One person, not two	Pass
RC-03	A list stored in the older format	Read as written	Pass
RC-04	A corrupted stored value	Read, not an error	Pass
RC-05	Correct title, authors and year together	One write to every passage	Pass
RC-06	Correct one field only	The others unchanged	Pass
RC-07	Correction with a blank title	Refused	Pass
RC-08	Correction with a blank year	Accepted	Pass
RC-09	A year that is not a year	Refused	Pass
RC-10	An author list of 200 names	Refused	Pass
RC-11	Correction to an absent document	404, nothing written	Pass
RC-12	Author line typed with commas	Split into names	Pass
RC-13	Author line typed with semicolons	Commas kept inside names	Pass
RC-14	Key from surname, year and title word	`vaswani2017attention`	Pass
RC-15	A paper with no usable metadata	Still citable	Pass

ID	Case	Expected	Result
RC-16	Two papers producing one key	Suffixed as BibTeX does	Pass

2.5 Capability, Models and Release

Table 2.5: Capability, model management and release test cases

ID	Case	Expected	Result
CM-01	Model sized against available, not installed, memory	Correct budget	Pass
CM-02	Prompt exceeding the loaded context	Map-reduce fallback	Pass
CM-03	Apple Silicon, no CUDA reported	GPU still used	Pass
CM-04	Vulkan device present, vendor tool absent	Advice still shown	Pass
CM-05	Task routed to an assigned model	Correct model	Pass
CM-06	Assigned model too large for free memory	Falls back, states why	Pass
CM-07	Version lower than published	Refused	Pass
CM-08	Release published as a draft	Detected and republished	Pass
CM-09	Update where the platform cannot prompt	Not installed	Pass

2.6 Interface

Table 2.6: Interface test cases

ID	Case	Expected	Result
UI-01	Press inside the page collapses the navigation	Collapses	Pass
UI-02	The collapse does not swallow the click causing it	Click delivered	Pass
UI-03	Reopening survives the next press	Stays open	Pass
UI-04	Automatic collapse resumes after navigating	Resumes	Pass
UI-05	Every feature resolves from its own path	All resolve	Pass

ID	Case	Expected	Result
UI-06	Nested path resolves to its parent feature	Resolves	Pass
UI-07	Unknown path	Falls back without error	Pass
UI-08	Workspace pages take the full window	Full width	Pass
UI-09	Update prompt with no confirmation available	Treated as not asked	Pass
UI-10	Pigments: earning records and applies	Correct	Pass

Chapter 3

TEST RESULTS

3.1 Suite Summary

Table 3.1: Automated suite results

Suite	Tests	Pass	Fail	Runtime
Backend (pytest)	1 042	1 042	0	~13 s
Interface (vitest)	213	213	0	~5 s
API contract	5	5	0	<1 s
Route reachability	16	16	0	<2 s
Total	**1 276**	**1 276**	**0**	

3.2 Metadata Extraction Accuracy

Two measurements, and the difference between them is the finding.

Table 3.2: Extraction accuracy: chosen versus randomly sampled papers

Field	Curated (18)	Random (56)
Title exactly correct	100%	92.9%
Author list exactly correct	100%	85.7%
Year exactly correct	100%	92.9%
All three simultaneously	100%	76.8%

The curated set was chosen by the authors. The random set was drawn across 15 arXiv subject classifications through the public interface, and scored against the archive's own catalogue records, so neither the papers nor the expected answers were selected here.

The first random measurement was **62.5%** on all three fields. The whole difference was three typesetting conventions that had not been examined:

1. a physics template spacing a centred author line widely enough to be indistinguishable from column separation, so two authors read as five fragments and none as a name;
2. a template running from the affiliations into the abstract with no heading, so the search for authors continued into the body and collected section headings;
3. a margin threshold added to discard an archive stamp, which was also discarding the first line of any centred title wide enough to begin left of it.

Of the 13 remaining failures, **three are not extraction errors**: the page reads "M. GRENDÁR" and "Audrey Quessadda-Vial" where the archive record disagrees with the paper itself.

3.3 Ablation

To establish which rules carry the result, each was disabled and the corpus re-scored.

Table 3.3: Rule ablation on the curated set

Configuration	Exact	Failure mode
All rules	7/7	:
No column splitting	1/7	38 names lost entirely
No affiliation word list	4/7	Employers admitted; no name lost
No acronym rule	6/7	“Google AI Language” admitted

Splitting rows into cells is the load-bearing rule. The word list only ever admits extra rows and never loses a name, which is why it remains a tiebreaker rather than the mechanism.

3.4 Performance

Table 3.4: Ingestion profile, one 40 kB paper

Stage	Time	Share
Text extraction	6.0 ms	0.2%
Metadata extraction	1.1 ms	0.04%
Chunking	0.2 ms	0.01%
Embedding and search	3073.8 ms	99.7%

This measurement prevented a rewrite. Everything a systems-language rewrite would have touched is 0.3% of runtime, and the remaining 99.7% is already compiled native code.

Chapter 4

DEFECTS FOUND AND FIXED

4.1 By Severity

Table 4.1: Defects found during the testing phase

#	Defect	Severity	Status
D-01	Metadata wrong on every published paper tested	Critical	Fixed; 92.9% on unseen papers
D-02	Released version 1.0.0 did not start	Critical	Fixed; packaged validation added
D-03	Model fallback unreachable (guard tested for empty, failure was wrong)	High	Fixed; plausibility check
D-04	Summarisation prompt exceeded its own context window	High	Fixed
D-05	Apple Silicon pinned to CPU	High	Fixed
D-06	Release published as a draft; testers told they were current	High	Fixed; publish asserted
D-07	Graphics advice hidden from AMD/Intel/open-driver machines	High	Fixed
D-08	Sidebar collapse swallowed the click that caused it	Medium	Fixed
D-09	Update prompt read a missing confirmation as "declined"	Medium	Fixed
D-10	Interface panel displayed a model that served nothing	Medium	Fixed; reads the routing table
D-11	Two dashboard figures never displayed a value in any release	Medium	Fixed
D-12	Panes sized by a guessed constant; broke when a heading was removed	Medium	Fixed
D-13	Contract test validated a directory that was about to be deleted	Medium	Fixed; parameterised
D-14	Figure silently dropped while the compile reported success	Medium	Fixed; PDF inspected
D-15	Ligatures and accents reached the search index	Low	Fixed

4.2 The Recurring Defect Class

Five of the defects above, D-01, D-03, D-06, D-07 and D-11, are the same mistake: *a check that tests for silence, when the failure mode is confident wrongness.* A guard asking whether a title is empty cannot catch a title that is wrong. A release asking whether a job failed cannot catch a job that succeeded into a draft. A dash meaning "nothing yet" cannot be distinguished from a dash meaning "the request failed".

The correction is the same in each case: test for the shape of a right answer rather than for the absence of a wrong one.

Chapter 5

STATUS AND OUTSTANDING WORK

5.1 A Defect No Test Environment Could Have Caught

One defect from the citation work is recorded for the method rather than the cause, because it changes what the interface suite can be relied on to prove.

The citation list did not appear. Every automated test passed, including tests that type into an editor and assert that the list opens with the expected rows, because those tests run in a simulated document object model that computes no layout: every element it reports is at the origin and has no size. Driving the real application and reading the element's position back showed the list present, populated with all twenty papers, and drawn 1382 pixels down a window 1000 pixels tall, inside a region that hides rather than scrolls what exceeds it. A second cause was found the same way: the compiled document displayed beside the editor is drawn above the list and concealed the column holding the citation keys.

The consequence for this document is narrow and should not be overstated. The interface suite proves that components mount, that state changes as expected, and that the calls they make exist. It cannot prove that anything is visible. Screenshots taken from the running application, and the platform verification in Section 5.2, are the only evidence of that, which is part of why the outstanding items below are treated as blocking rather than cosmetic.

5.2 Current Status

All automated suites pass: 1 276 tests across four layers. The application has been released publicly through fifteen versions, and the packaged installer has been validated on Linux by running it.

5.3 Outstanding Verification

This document is issued now because the testing phase is due, and it is **provisional in two specific respects**. Both are stated rather than omitted.

Table 5.1: Verification not yet performed

Outstanding	Why, and what is required
Cross-platform verification of the replaced interface	The interface was replaced after the last full platform sweep. The automated suites pass and the application has been run on Linux, but Windows and macOS have not been re-verified against the new interface. Required: install the packaged build on each and run the ingest, map, write and compile paths.

Outstanding	**Why, and what is required**
Scribe on all platforms	The writing and compilation path depends on the bundled typesetting engine and on filesystem behaviour that differs between platforms. It is verified on Linux. Required: compile a document containing a figure, a table and an index on each platform, and confirm the PDF contains them.
Citations on all platforms	Verified on Linux, including a document compiled end to end with numbered references. Two platform-specific risks remain. The bundled engine runs the bibliography processor itself while a build from source does not, so the two paths differ and only one has been exercised on each platform. And the citation list is positioned from measurements taken in the browser engine, which is a different engine on each operating system. Required: cite a paper and compile on each platform, and confirm the list is drawn where the cursor is.

Reporting these as complete would be the same defect this document spends a chapter on: claiming a result from the absence of a failure that was never looked for.

5.4 Revision

This document will be revised once the verification in Section 5.2 is complete. The suite figures in Chapter 3 are current as of the date of issue and are reproducible by running the suites named beside them.